

Project Report

Title: Airbnb Rental Price Prediction Using Machine Learning

1. Introduction

Airbnb is a global platform that connects hosts offering unique stays with guests seeking accommodation. One of the key concerns for hosts is setting the right price for their property listings. Pricing too high may deter customers, while pricing too low can lead to loss of potential revenue. To address this issue, the project leverages machine learning to predict the optimal rental price for Airbnb listings in New York City using available listing data such as location, room type, number of bedrooms, and other features.

2. Abstract

The goal of this case study is to build a predictive model that estimates Airbnb rental prices in NYC. Using a dataset that includes detailed information on Airbnb listings, the project performs data cleaning, exploratory data analysis (EDA), and feature engineering. Several machine learning models including Linear Regression, Decision Trees, and Random Forest Regressors are trained and evaluated based on performance metrics such as RMSE and R^2 . The final model is chosen based on its predictive accuracy and generalization capability. The project provides insights into key factors affecting rental prices and serves as a decision-support tool for Airbnb hosts.

3. Tools Used

Programming Language: Python

Libraries & Frameworks:

- Data Handling: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Modeling: Scikit-learn

Platform: Jupyter Notebook

4. Steps Involved in Building the Project

1. Data Collection:

- Dataset loaded from a CSV file containing listings data for New York City.

2. Data Cleaning and Preprocessing:

- Removed irrelevant or duplicate columns.
- Handled missing values and outliers.
- Converted categorical variables into numerical form using one-hot encoding.

3. Exploratory Data Analysis (EDA):

- Visualized distributions of prices, room types, boroughs, etc.
- Correlation analysis performed to identify key influencing factors.

4. Feature Engineering:

- Derived new features (e.g., price per room, room availability ratio).
- Scaled and normalized data for certain models.

5. Model Building:

- Applied multiple regression algorithms including:
 - Linear Regression
 - Decision Tree Regressor
 - Random Forest Regressor
- Split data into training and test sets (e.g., 80/20).

6. Model Evaluation:

- Evaluated models using metrics such as:
 - RMSE (Root Mean Square Error)
 - MAE (Mean Absolute Error)
 - R^2 Score
- Random Forest achieved the best performance among all tested models.

5. Conclusion

This project successfully demonstrates how machine learning can be utilized to predict Airbnb rental prices in NYC. Random Forest Regressor provided the most accurate predictions by effectively handling non-linearity and feature interactions. The analysis highlights that location, room type, and number of reviews are significant factors influencing rental prices. Future enhancements could include integrating external data (e.g., seasonal trends, events) or deploying the model as a web application for real-time price recommendations.